MMA3001 Project

Limited-Observation Calibration of Composite Beam Compliance Models

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Abstract

This project evaluates whether a model trained on finite-element-analysis (FEA) responses can be calibrated using two observations and predict unseen responses for a different composite material or layup. A completed linear-static cantilever sweep provides 80 training cases and 80 target cases across material interpolation, material extrapolation, layup transfer, and geometry extrapolation. A mechanics-informed compliance regression, quadratic response surface, and Gaussian process are compared using five-fold training cross-validation and three calibration-pair repeats per target set. Calibration reduced material-interpolation held-out MAPE from approximately 50% to below 0.2% for all three models. Material-extrapolation results exposed a non-identifiable calibration pair, while calibration offered little benefit when nominal predictions were already accurate. The study demonstrates FEA-to-FEA model transfer and the importance of informative observations; it does not establish physical landing-gear accuracy. The physical interpretation of the linear solution is limited for cases with high displacement-to-length ratios.

1 Introduction

Composite landing-gear design can require costly physical testing to determine how geometry, layup, and laminate stiffness affect deflection. This project investigates whether a model trained on FEA results can be corrected with only a few observations and used to predict unobserved responses. The long-term motivation is to apply the same workflow to physical test observations when they become available.

The present trial is a controlled methodology demonstration, not a landing-impact simulation or a mass optimization. It uses a straight cantilever as a tractable benchmark. No physical testing observations are included, and the material cards are numerical assumptions. Consequently, the validation performed here compares model predictions with held-out FEA results generated using the same modeling framework; it does not establish agreement with a manufactured gear.

2 Finite-element model

The beam is a flat rectangular layered shell with fixed length $L = 0.25\,\mathrm{m}$. The axis is global X, width is global Y, and the applied transverse force is global $-Z$. The root edge is fully fixed. A total 10 N load is distributed equally across nodes on the free tip edge. The completed sweep used linear-static analysis with orthotropic linear-elastic material cards. Geometric stiffness changes due to deformation were not included.

The mesh uses SHELL181 layered shell elements [1], with 40 divisions along the length and 6 across the width. Three through-thickness integration points are assigned per ply. This is a two-dimensional shell discretization with a layered section, not a three-dimensional solid model. All cases use the same mesh.

Dataset A uses a symmetric twill/glass stack built by repeating carbon twill (T) and glass (G) in the half-stack and mirroring about the mid-plane. Carbon twill plies are 0.20 mm and glass plies are 0.10 mm.

Dataset B's alternate-layup set uses a symmetric repeating carbon UD (U), carbon twill (T), and glass (G) half-stack, truncated to the needed half-ply count and mirrored; both carbon ply types are 0.20 mm and glass is 0.10 mm.

Table 1: Stack thicknesses used in the two layup families.

Ply count	Twill/glass total thickness	UD/twill/glass total thickness	Unit
8	1.2	1.4	mm
16	2.4	2.8	mm
24	3.6	4.0	mm
32	4.8	5.4	mm

Orthotropic elastic and shear properties are specified in the example configuration. For the transfer experiment, a carbon factor scales the elastic and shear moduli of carbon plies, and a glass factor scales those of glass plies; density, Poisson ratios, and ply thickness do not change when these factors vary. The cards and cured ply values are computational assumptions, not measured project material properties.

The response is tip compliance,

$$C = \frac{|\bar{u}_{z,\mathrm{tip}}|}{P}, \tag{1}$$

where $\bar{u}_{z,\mathrm{tip}}$ is the mean vertical tip-edge displacement and $P = 10\,\mathrm{N}$. Compliance has units of m/N. For the linear-static formulation, compliance is independent of applied load within the mathematical model. The archived case files contain the response, geometry, layup definition, mass, and calculated equivalent section properties.

2.1 Inputs, outputs, and supported domain

The configuration is a JSON file with SI units. Beam length, load, width, ply thickness, material moduli, and density must be positive and finite. The present training design uses a 0.25 m beam, 10 N load, widths 0.02–0.10 m, ply counts 8, 16, 24, and 32, and carbon/glass scale pairs $(1, 1)$, $(1, 2)$, $(2, 1)$, and $(2, 2)$. Target geometry cases use widths 0.015–0.15 m and even ply counts up to 100. Only symmetric `tw_glass` and `ud_tw_glass` stacks made of zero-degree plies are supported. The output is mean vertical tip displacement in metres and compliance in m/N; the CSV also records mass, calculated section stiffnesses, design identifiers, material-scale factors and target-set labels.

The experiment is restricted to straight rectangular beams, the two specified symmetric zero-degree layup families, and the configured material cards. It does not cover arbitrary ply orientations, damage, contact, or load-dependent material behavior. The completed data and model outputs are archived under `results/property_transfer_extended/`. The report describes that completed experiment; later exploratory code is not used as evidence.

3 Experiment design and data

3.1 Dataset A: model training

Dataset A contains the full 5 by 4 by 4 factorial design: five widths, four ply counts, and four carbon/glass property-factor combinations. The combinations are $(1, 1)$, $(1, 2)$, $(2, 1)$, and $(2, 2)$.

- Widths: 0.02, 0.04, 0.06, 0.08, and 0.10 m.
- Ply counts: 8, 16, 24, and 32.
- Carbon/glass stiffness factor pairs: $(1, 1)$, $(1, 2)$, $(2, 1)$, and $(2, 2)$.

This gives 80 FEA responses. The training cases span the base twill/glass stack and the configured carbon/glass stiffness variation.

3.2 Dataset B: target sets

Dataset B contains four 20-case sets (80 cases total). Sets 1–3 use the Dataset A widths and ply counts; geometry extrapolation uses its separately defined levels. Together, Datasets A and B comprise 160 completed analyses.

Table 2: Dataset B target design.

Target set	Change from Dataset A
Material interpolation	Twill/glass layup; carbon/glass factors $(1.5, 1.5)$.
Material extrapolation	Twill/glass layup; factors $(2.25, 0.75)$.
Layup transfer	Symmetric UD/twill/glass layup; factors $(1, 1)$.
Geometry extrapolation	Twill/glass layup; widths 0.015, 0.04, 0.06, 0.08, 0.15 m and ply counts 8, 16, 32, 100; factors $(1, 1)$.

The layup-transfer case is important to interpret correctly. Its FEA targets were generated with the material cards in the configuration at factors $(1, 1)$. Those same configured cards and the known alternate ply sequence were used to calculate nominal laminate section properties for prediction. Thus nominal predictions were not made with no material knowledge: this set tests transfer to a different layup using known nominal material cards. During calibration, the model was allowed to adjust the carbon and glass factors around that baseline using only response observations. The configured truth factors were not supplied to the calibration optimizer.

4 Models, calibration, and validation

4.1 Section-property predictors

For each case, the effective laminate bending rigidity per unit width D_{eff} is calculated from the laminate stiffness matrices [2]. The transverse-shear rigidity per unit width is approximated as

$$S_{\text{eff}} = \sum_k G_{xz,k} t_k. \tag{2}$$

For a case of width b, the bending and corrected shear rigidities are $EI = bD_{\text{eff}}$ and $kGA = \kappa b S_{\text{eff}}$, with $\kappa = 5/6$. The mechanical basis terms are

$$x_b = \frac{L^3}{3EI}, \qquad x_s = \frac{L}{kGA}. \tag{3}$$

4.2 Model families

Three models are fit to Dataset A only:

1. **Mechanics-informed compliance regression:** nonnegative, no-intercept least squares in x_b and x_s, $\widehat{C} = \beta_b x_b + \beta_s x_s$.
2. **Quadratic response surface:** second-order polynomial in standardized logarithms of width, total thickness, D_{eff}, and S_{eff}, fit to log compliance with ridge regularization.
3. **Gaussian process:** Matérn 5/2 kernel plus a white-noise term, using the same log predictors and log compliance.

Five-fold cross-validation on Dataset A estimates within-training-domain predictive performance. Target FEA responses are not used to fit or select the Dataset A models. The mechanics-informed bending/shear basis represents the dominant compliance contributions of the linear beam benchmark.

4.3 Calibration and held-out validation

For each of the four target sets, three calibration repeats are performed. A pair of target cases is randomly selected for each repeat; the same pair is used across model families. For each fitted model, nonlinear least squares estimates two multipliers, q_C and q_G, by minimizing the squared log-compliance residuals at the selected cases:

$$(q_C^*, q_G^*) = \arg\min_{q_C, q_G} \sum_{i \in \mathcal{C}} \left[\log \widehat{C}_i(q_C, q_G) - \log C_{i,\mathrm{FEA}}\right]^2, \qquad |\mathcal{C}| = 2. \tag{4}$$

Candidate factors are applied to the configured ply material cards, after which D_{eff}, S_{eff}, EI, and kGA are recalculated. The trained model coefficients or surrogate surface are held fixed. The compliance observations are the calibration data; calculated section properties are model inputs, not target observations.

The other 18 cases are held out for validation in each repeat. The calibration cases do not appear in the validation metrics. Reported measures include MAPE, maximum absolute percentage error, RMSE, and R^2. Calibration fits two unknown factors from the algebraic minimum of two observations, so a poorly chosen pair may produce non-identifiable or unstable factors. Jacobian rank and condition number are recorded.

5 Results and model evaluation

The results below are from the completed 160-case linear-static experiment. Dataset A was used for fitting and cross-validation; target responses were reserved for two-observation calibration and held-out evaluation.

5.1 Dataset A cross-validation

Table 3: Five-fold cross-validation scores on Dataset A.

Model	MAPE (%)	RMSE (m/N)	R^2
Mechanics-informed	0.1083	4.364×10^{-6}	0.9999994
Quadratic surface	0.0053	2.797×10^{-7}	0.999999998
Gaussian process	0.0195	2.171×10^{-6}	0.9999999

The quadratic surface had the lowest cross-validation error over the sampled Dataset A grid. The mechanics-informed model is more directly interpretable, while its error was higher. These results do not themselves establish target-set or physical accuracy.

5.2 Target-set held-out performance

Table 4 gives the mean MAPE across the three repeats, with standard deviation across repeats. Each MAPE is calculated over that repeat's 18 held-out cases.

For the same-layup material-interpolation set, nominal predictions assume factors $(1, 1)$ while the target uses $(1.5, 1.5)$. Calibration recovers factors near the target and lowers held-out MAPE from about 50% to below 0.2% for all models. For material extrapolation, nominal MAPE is about 99%; informative calibration pairs again reduce it substantially. However, one random pair (case IDs 37 and 39) consists of two 32-ply cases at widths 0.06 and 0.10 m. Width changes both bending and shear compliance by a similar scale, so that pair does not separate carbon and glass effects. The mechanics Jacobian is rank deficient (rank 1; condition number about 2.2×10^{16}), and other models also produce poor scale estimates for that repeat. The three-repeat average therefore hides a calibration-design failure; the fitted factors and conditioning must be examined alongside prediction scores.

Table 4: Held-out MAPE before and after calibration, mean ± standard deviation across repeats.

Target set	Model	Nominal MAPE (%)	Calibrated MAPE (%)
Material interpolation	Mechanics-informed	50.150 ± 0.008	0.125 ± 0.061
	Quadratic	50.000 ± 0.001	0.006 ± 0.001
	Gaussian process	49.974 ± 0.003	0.011 ± 0.002
Material extrapolation	Mechanics-informed	98.805 ± 0.409	1.007 ± 1.553
	Quadratic	98.619 ± 0.422	0.967 ± 1.656
	Gaussian process	98.586 ± 0.422	0.947 ± 1.622
Layup transfer	Mechanics-informed	0.067 ± 0.005	0.122 ± 0.078
	Quadratic	0.204 ± 0.002	0.066 ± 0.036
	Gaussian process	0.081 ± 0.002	0.060 ± 0.010
Geometry extrapolation	Mechanics-informed	0.254 ± 0.009	0.305 ± 0.155
	Quadratic	0.283 ± 0.032	0.359 ± 0.102
	Gaussian process	0.159 ± 0.021	0.246 ± 0.090

5.3 Dataset B subset 3: alternate-layup detail

Subset 3 is the 20-case `layup_transfer` set, IDs 40–59. Its widths are 0.02–0.10 m and ply counts are 8, 16, 24, and 32. All FEA target cases have configured material factors $(1, 1)$. Dataset A is the only training data; subset 3 is used solely for calibration and validation.

Table 5: Subset 3 calibration pairs, fitted scale factors, and held-out MAPE.

Model	Repeat	Calibration IDs	q_C	q_G	Nominal MAPE (%)	Calibrated MAPE (%)
Mechanics	1	48, 55	1.0022	0.9736	0.0719	0.0877
Mechanics	2	54, 56	1.0065	0.9195	0.0668	0.2112
Mechanics	3	54, 58	1.0003	1.0178	0.0625	0.0669
Quadratic	1	48, 55	0.9965	1.0241	0.2062	0.0505
Quadratic	2	54, 56	0.9946	1.0483	0.2046	0.1070
Quadratic	3	54, 58	0.9975	1.0000	0.2024	0.0395
Gaussian process	1	48, 55	0.9985	1.0082	0.0829	0.0481
Gaussian process	2	54, 56	0.9980	1.0110	0.0800	0.0648
Gaussian process	3	54, 58	0.9981	1.0071	0.0790	0.0659

The corresponding validation IDs are: repeat 1, 40–47, 49–54, 56–59; repeat 2, 40–53, 55, 57–59; repeat 3, 40–53, 55–57, 59. Thus, each repeat validates on 18 cases and excludes only its own calibration pair. Fitted factors remain near $(1, 1)$, consistent with the configured target cards. Nominal predictions were already accurate because the model received those known cards and the alternate layup's calculated effective properties. Calibration modestly improved quadratic and Gaussian-process average errors but worsened the mechanics-informed average. This is not a test of estimating entirely unknown material properties: it is a layup-transfer test with known nominal material cards and optional response-based adjustment.

6 Discussion and limitations

The results show that a small number of observations can correct a response model when the target variation is represented by the calibration parameters and the selected observations are informative. They also show that low Dataset A cross-validation error does not guarantee robust target calibration. Calibration pair selection matters: two response measurements should produce distinct sensitivities to the unknown

property factors. A high Jacobian condition number or deficient rank signals that the fitted factors should not be interpreted.

The alternate-layup and geometry-extrapolation cases had low nominal MAPE because their configured section properties were available and close to the training response domain. Calibration brought little benefit there and sometimes increased error by adjusting the property factors to absorb small model residuals. This is consistent with a correction method that should be applied only when justified by observations and diagnostics.

The linear-static sweep had a maximum target $|\delta|/L$ of 1.70; 21 of 80 target cases and 22 of 80 training cases exceeded 0.1. The surrogate scores quantify agreement with computed linear FEA responses, but those extreme displacements cannot be interpreted as physically valid small-deflection predictions at 10 N. Reducing the load would reduce displacement while preserving ideal linear compliance; restricting the geometry domain is another option. These limitations prevent a claim that the reported sub-percent surrogate errors represent physical gear accuracy.

All cases used a 40 by 6 mesh. The completed result archive does not establish mesh convergence for every transfer configuration. Mesh sensitivity, reaction balance, element orientation, support fidelity, load application, and possible locking must be verified before interpreting small model errors as engineering accuracy. The study does not address impact dynamics, laminate failure, progressive damage, manufacturing constraints, or curved landing-gear geometry.

7 Conclusion and future work

The completed sweep demonstrated a training-calibration-validation workflow for composite cantilever compliance and compared three model families. The quadratic surface achieved the lowest Dataset A cross-validation MAPE (0.0053%). Two-observation calibration substantially improved predictions for the same-layup material-transfer sets, provided the observations supplied independent information. It sometimes worsened predictions when nominal section properties already described the target well. The alternate-layup set used known nominal material cards, so it did not demonstrate prediction without a baseline material description.

Future work should select calibration pairs using sensitivity separation, increase the number of calibration repeats, and assess robustness to measurement noise. Representative mesh-convergence and reaction-balance checks should accompany a restricted or lower-load linear benchmark. A large-deflection formulation could then extend the method beyond small-deflection kinematics, with fresh training and validation data. Physical observations, curved landing-gear geometry, and mass/stroke optimisation are subsequent applications requiring their own verification and validation.

8 AI-use reflection

Generative AI was used as a development assistant during planning, code changes, debugging, and report drafting. It helped structure the experimental design, explain the calibration procedure, identify inconsistencies between the intended and implemented sweep, and prepare documentation.

AI-generated code and explanations were checked against the repository, unit tests, the project requirements, and MAPDL output where available. Unit tests verify design construction and command setup, but they do not verify the physical correctness or convergence of a MAPDL solution. A limitation of AI assistance was that generated interpretations could sound conclusive despite incomplete evidence. The report was checked against the completed sweep and evaluation outputs, and revised to distinguish surrogate agreement with FEA from physical validation.

References

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[2] J. N. Reddy, *Mechanics of Laminated Composite Plates and Shells: Theory and Analysis*, 2nd ed., CRC Press, 2004.

[3] Ansys, Inc., *PyMAPDL documentation*, Ansys. https://mapdl.docs.pyansys.com/version/stable/